

DIGITAL FORENSICS

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References

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- Digital Forensics and Incident Response, 4th Edition,
 - Gerard Johansen, Packt, 2025
- Roadmap of digital forensics investigation process with discovery of tools, Cyber Security and Digital Forensics,
 - Anita Patil, et al. (2022):241-269.
- SWGDE 18-F-001-2.0 / SWGDE 18-F-002-2.0 /
- SWGDE 17-F-002-2.1 / SWGDE 23-Q-001-1.1 /
- NIST SP 800-61 Rev. 3
- IETF RFC 3227
- <https://www.sans.org/blog>

DAY 2

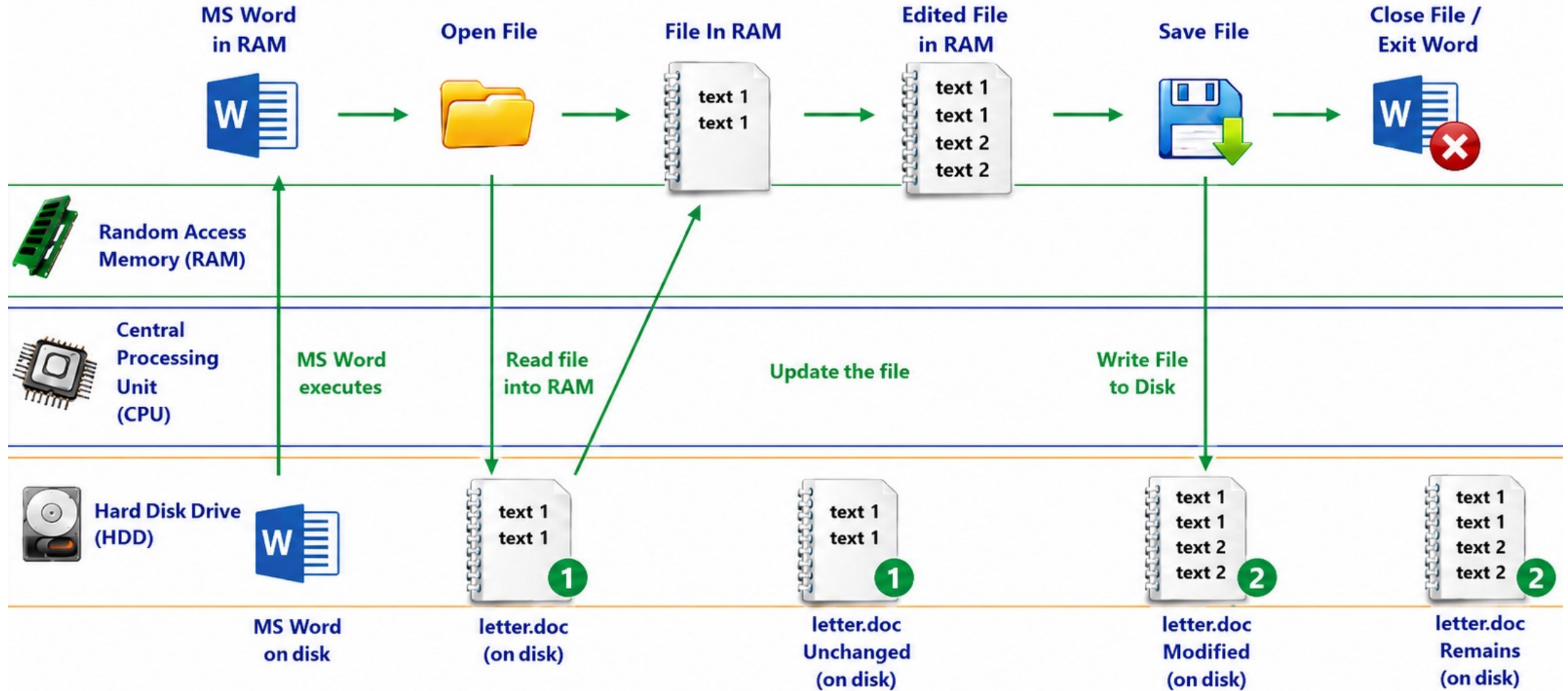
**ACQUISITION OF THE DIGITAL
EVIDENCE**

National Software Reference Library (NSRL)

- National Software Reference Library (NSRL)
- <https://www.nist.gov/itl/csd/secure-systems-and-applications/national-software-reference-library-nsrl>
- NSRL is a NIST-maintained collection of cryptographic hashes for known software files.
- Investigators use its **Reference Data Set (RDS)** to identify and filter out common, trusted files, reducing the amount of data that must be examined.
- Autopsy has an option to add RDS.

How Computer Works

How Computer Works



Memory Artifacts

Memory Artifacts

- **Physical Memory (RAM)** – Contains live processes, network connections, credentials, encryption keys, and other volatile data.
- **Page File (pagefile.sys)** – Stores memory pages moved from RAM to disk and may contain fragments of processes, documents, credentials, and deleted data.
- **Hibernation File (hiberfil.sys)** – Saves memory contents when Windows hibernates and may preserve a near-complete snapshot of RAM.
- **Crash Dumps** – Memory captured after a system or application crash, such as 'MEMORY.DMP' or minidump files.
- **Virtual Machine Memory Files** – Files such as .vmem , .vmss that store the memory state of a virtual machine.

Physical Memory (RAM)

- Running processes – Programs and active services.
- Network connections – Active or recently used IP, ports, sessions.
- Loaded DLLs and drivers used by processes.
- Command history – executed commands in PowerShell, CMD
- Logged-in users and sessions
- Malware traces – Injected code, hidden processes, hooks, and suspicious memory regions.

Physical Memory (RAM)

- Encryption keys and credentials – passwords, keys tokens
- Open files and handles – Files, registry keys, sockets, and other resources accessed by processes.
- Clipboard and application data
- System information – Operating system details, uptime, time, and memory configuration.

Windows Artifacts

Windows Artifacts

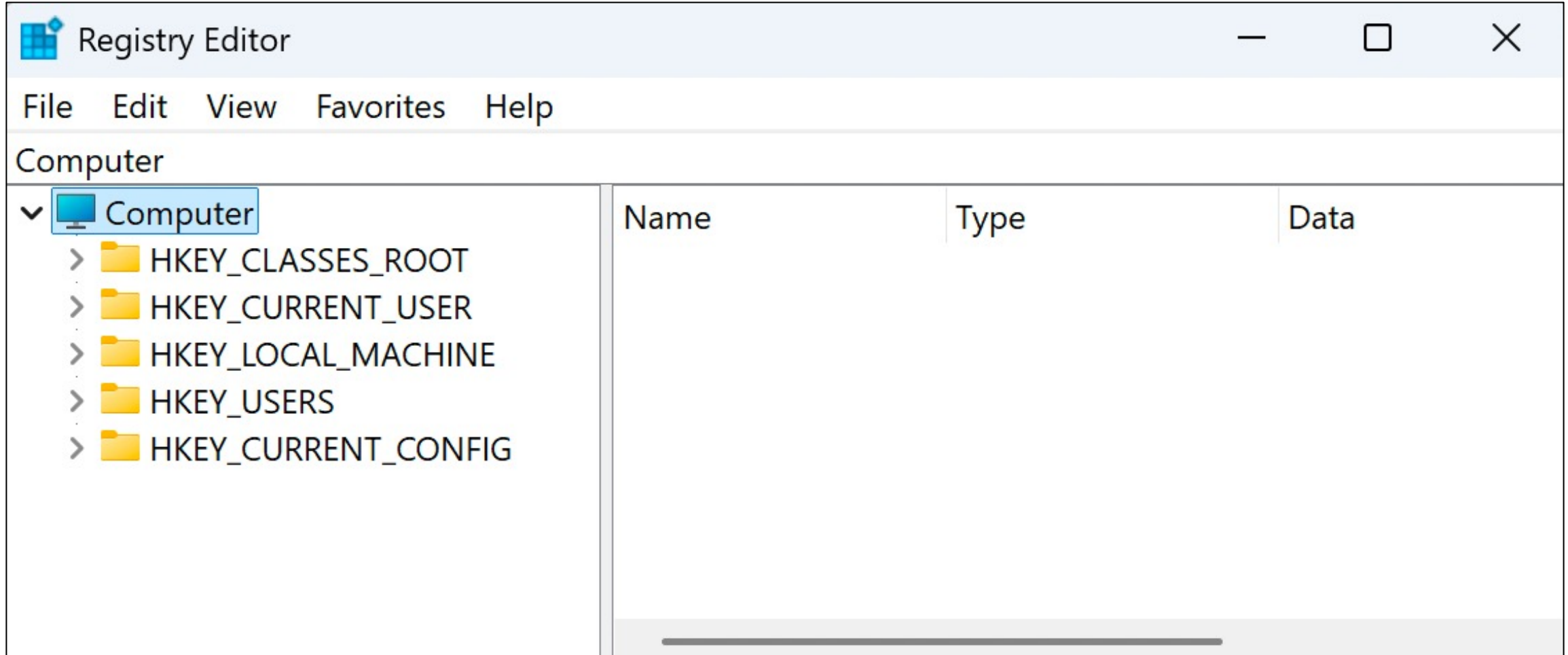


- System Information
- Application Execution
- Account Usage
- Cloud Storage
- Network Activity and Physical Location
- File and Folder Opening
- Deleted Items and File Existence
- External Device/USB Usage
- Browser Activity

Windows Forensic Analysis
— P O S T E R —

Available in **SIFT** Desktop

Windows Registry



<https://learn.microsoft.com/en-us/windows/win32/sysinfo/predefined-keys>

Windows Registry – Files and Locations

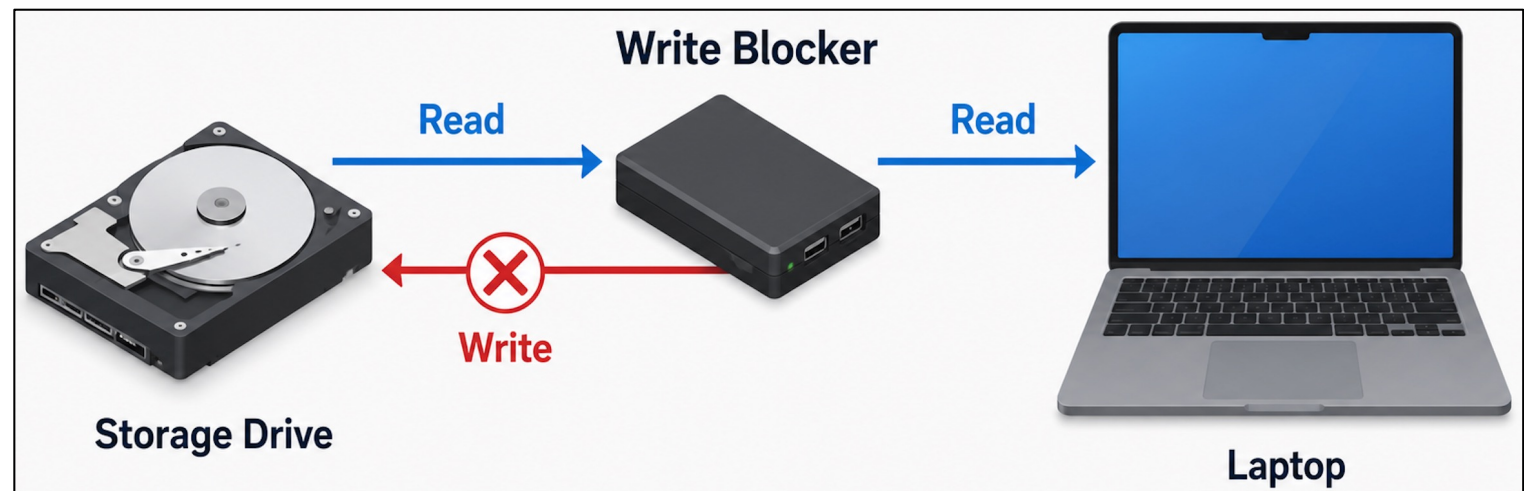
Registry File	Default Location	Purpose
NTUSER.DAT	C:\Users\<user-account>\NTUSER.DAT	Stores settings and preferences for an individual user.
DEFAULT	C:\Windows\System32\config\DEFAULT	Stores settings used by the Windows logon environment and system processes.
SAM	C:\Windows\System32\config\SAM	Stores local user accounts, groups, and password hashes.
SECURITY	C:\Windows\System32\config\SECURITY	Stores local security policies, LSA secrets, and cached domain credentials.
SOFTWARE	C:\Windows\System32\config\SOFTWARE	Stores Windows and installed-software configuration.
SYSTEM	C:\Windows\System32\config\SYSTEM	Stores hardware, drivers, services, startup settings, and control sets.
System Profile NTUSER.DAT	C:\Windows\System32\config\systemprofile\NTUSER.DAT	Stores profile settings for the Windows Local System account.



Acquisition of The Digital Evidence

Acquisition Process

- The guiding principle is to **minimize changes** to the source data.
- This is usually accomplished by the use of **write-blockers**
- **Write-blockers**: hardware or software intended to allow reading data from a storage device without allowing changes (writes) to the source media.



Ai generated

Types of Acquisitions

- Dead Box / Static Acquisition (powered off)
 - Hardware tools
 - Required decryption key if encrypted
 - If damaged, requires advanced tools to extract
- Live Acquisitions (running machine)
 - Trusted binaries (tools) from controlled media
 - Requires admin permissions to access some OS files/directories

Advantages of Live Acquisitions

- Volatile system data in RAM can be acquired
 - Running processes, net connections, settings, decryption keys, removed files (slides 9-10)
- Mounted encrypted drives can be accessed
 - Mount encrypted drive = decrypt it and assign it a drive letter (D:, E:)

Acquisition Methods

- Disk-to-image file acquisition
 - Most common for full disk acquisition
- Disk-to-disk acquisition
 - Used when disk-to-image is not possible
- Logical/sparse acquisition
 - captures only specific files of interest to the case
 - collects fragments of unallocated (deleted) data

Acquisition Methods

- A Disk-to-Image acquisition is a bit-by-bit copy of the disk, **preserving** all readable sectors.
 - copying block/sector-level, **NOT files**
- This is why **Disk-to-Image** is the preferred acquisition method in digital forensics.
- FTK imager

Acquisition Considerations

- Time to perform the acquisition
- Where the evidence is located (remote vs local)
- Mounted encrypted disks
- Size of the source and enough space for destination storage
- Wiped destination storage drive (sterile media)
- Operating System & filesystem
- Reliability of the acquisition tools
- Digital signatures and evidence validation

Acquisition Tools

- Hardware
- Software

Acquisition – HW Tools

Jump Kits / Response Kits

- Forensic laptop/workstation, bootable media, write blockers
- Evidence storage drives
- Acquisition software, volatile-data collection tools
- Camera or secured mobile camera
- Hashing and integrity-verification tools
- Adapters and cables – SATA, IDE, USB-A, USB-C, NVMe, M.2, power adapters, card readers, network

Acquisition – HW Tools

Jump Kits / Response Kits

- Evidence documentation – Chain-of-custody forms, evidence inventory logs, acquisition worksheets, **incident notes**, labels, and permanent markers.
- Evidence packaging materials – Antistatic bags and suitable boxes
- Faraday bags – for isolation from cellular, Wi-Fi, Bluetooth, or remote-wipe commands.
- Basic hand tools – Nonmagnetic screwdrivers, pliers, tweezers, flashlight, magnifier, and tools for safely removing drives from devices.

Acquisition – HW Tools

Jump Kits / Response Kits



Acquisition – HW Tools

Disk Duplicator + Write-Blocker

Standalone imaging device

- built-in write blocker
- wipe output
- image/clone drives



<https://superimager.com>

Acquisition – SW Tools

OS / Forensics Suites

Forensic Linux Distributions

- SANS Investigative Forensic Toolkit (SIFT)
- CAINE
- Kali Linux Forensics Mode
- Bootable USB vs Virtual Machine

Acquisition and analysis Suites

- FTK Forensics Toolkit
- Autopsy
- X-Ways Forensics
- KAPE



SANS Investigative
Forensic Toolkit (SIFT)



CAINE (Computer Aided
INvestigative Environment)



Kali Linux
Forensics Mode



FTK Forensics Toolkit



Autopsy



X-Ways Forensics



dd



KAPE

Forensic Disk Image Formats

- RAW (dd)
 - A simple bit-by-bit copy of the source media. Widely supported but usually large and contains limited metadata.
- E01
 - EnCase forensic format. Supports compression, case metadata, segmentation, and integrity checks.
- AFF
 - Open forensic formats designed for flexible storage, compression, metadata, and large-scale evidence handling.

Mounting Acquired Images

- Windows
 - Arsenal Image Mounter
 - OSFMount
 - FTK Imager
- Linux
 - Built in `mount` Linux command

Remote Acquisition

- Network bandwidth constrains
- Full disk acquisition is challenging
- Key artifacts are collected through EDR.
- In Cloud infrastructure, large logs, pcaps can be analyzed remotely without downloading them.

Lab 2: Live System Acquisition